GoodPiggies Presentation

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Live Demo

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Design

OG Design Goals vs. New Design Goals

- Original: rule-agnostic engine + generic view layer
- How they evolved: build a world before level, richer rules, authoring env importance increased
- Openness and Flexibility:
  - Physics Engine: ✓
  - Components: —
  - View: ✓

Stable Design vs. Changed Design

Engine Api Boundary:

- api/ vs internal/ split
- Core interfaces (GameEngine, World)
- Factory Pattern for vehicle assembly
-

Authoring Env:

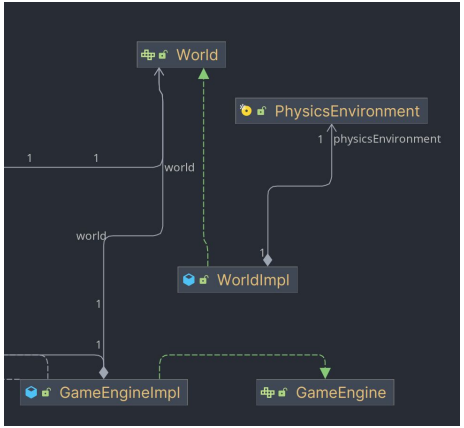
- Creating maps
  - Grid blocks ⟶ Splines
- Data Storage

3

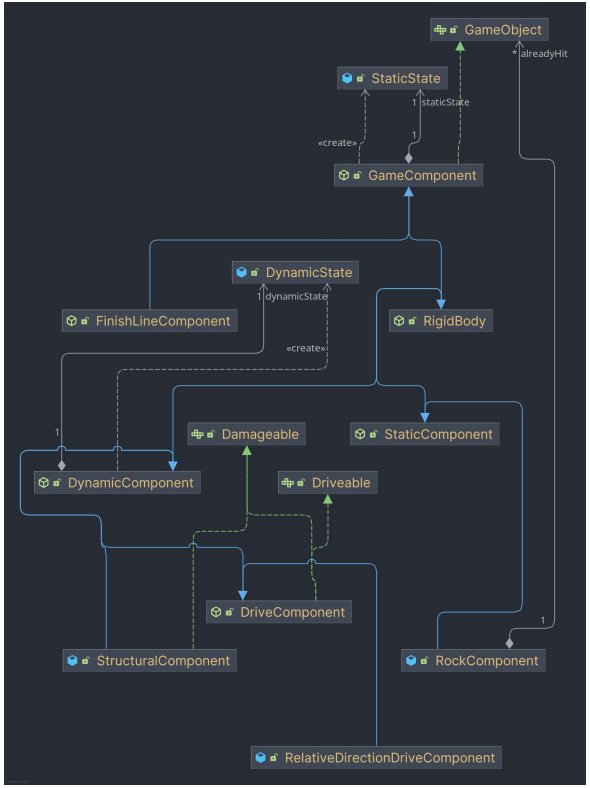
Notable APIs

Model + Engine API

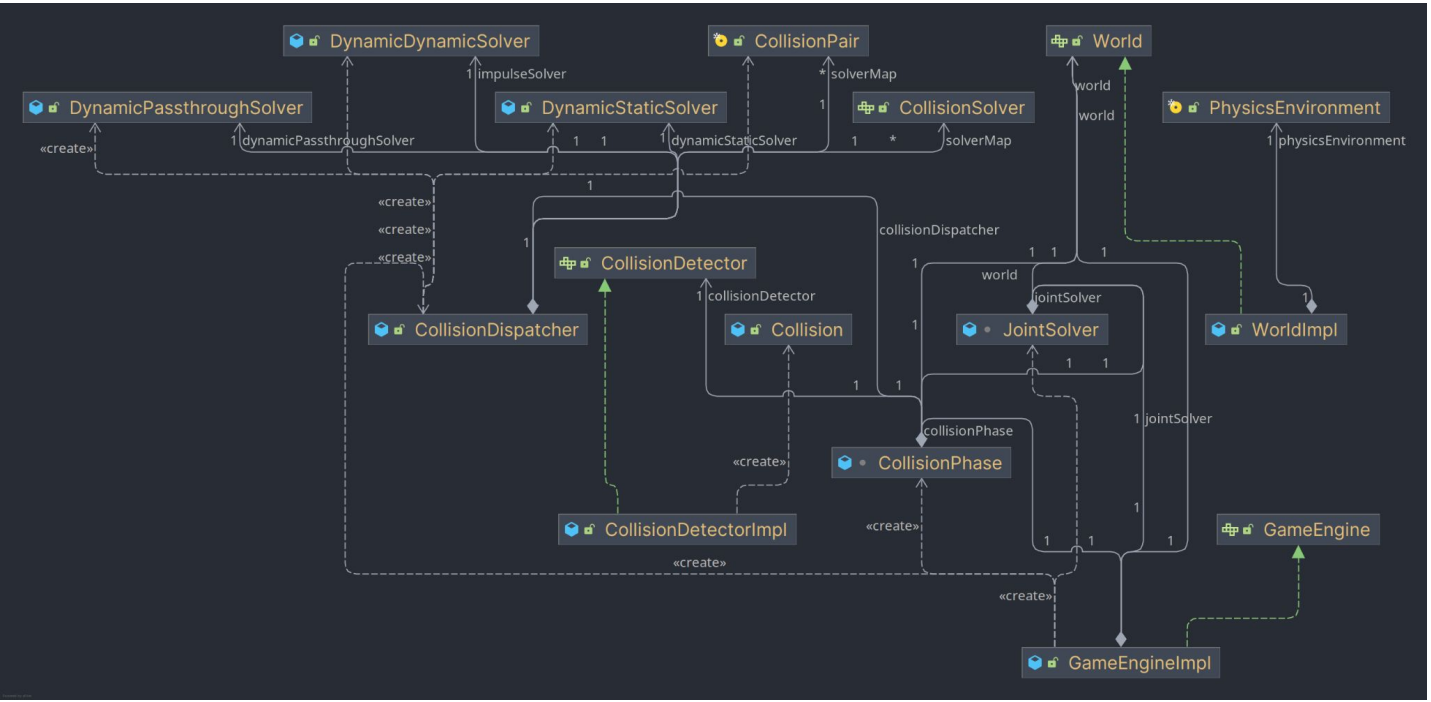
- Make adding components easy!
- Extensibility drove design of component representation, collision handling, and joint resolution.
- **`World`** & **`GameEngine`** tie everything together:



Model + Engine API: Components



Model + Engine API: Collisions





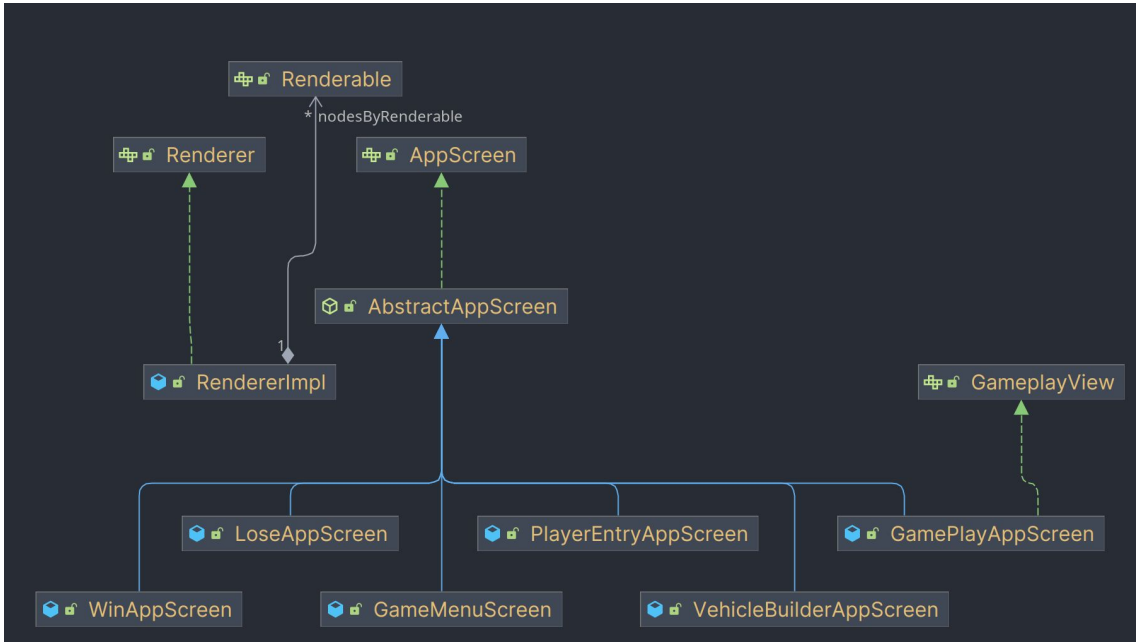
View APIs

- Public methods
- External services it depends on
- Well-designed code

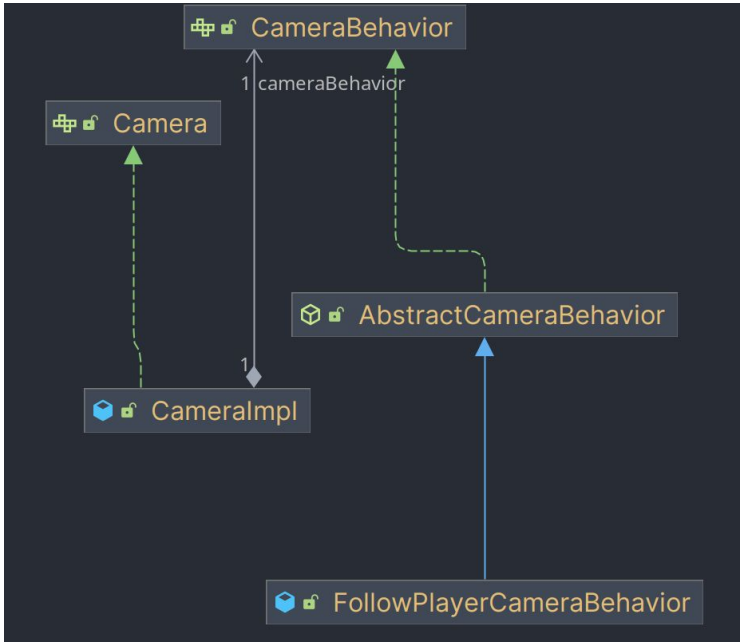
(screen management, camera abstraction, renderer)

- How it changed
- Use case example

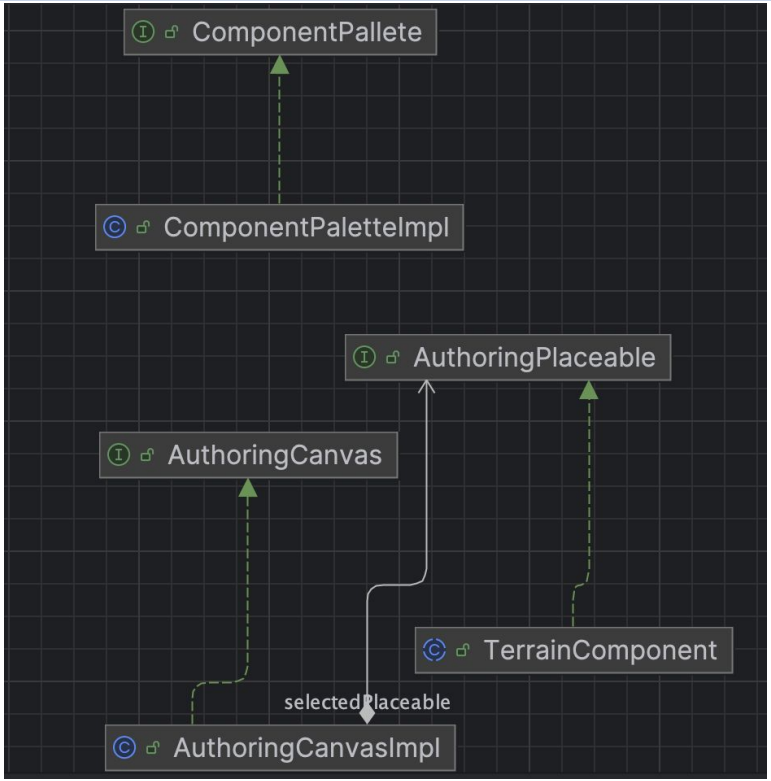
View APIs


Renderable
* nodesByRenderable
Renderer
AppScreen
AbstractAppScreen
1
RendererImpl
GameplayView
LoseAppScreen
PlayerEntryAppScreen
GamePlayAppScreen
WinAppScreen
GameMenuScreen
VehicleBuilderAppScreen

View APIs


CameraBehavior
1 cameraBehavior
Camera
AbstractCameraBehavior
1
CameraImpl
FollowPlayerCameraBehavior

View APIs


ComponentPallete
ComponentPaletteImpl
AuthoringPlaceable
AuthoringCanvas
TerrainComponent
selectedPlaceable
AuthoringCanvasImpl

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Teamwork

Wireframe - Sprint Planning vs. Reality

- Wireframe
- Points of difference from plan:
  - Authoring environment
  - Engine
  - Challenge features

Learnings from Agile/Scrum

- Emilio: Learned to be realistic about what you can get done.
- Georgiy: Learned to break issues/tasks down into smaller parts.
- Iurii: Learned that Agile is quite useful for tracking the overall picture
- Mason: Learned that feedback's value depends on how you receive it.
- Michael: Learned to make GitLab issues and how to assign story points.
- Paulina: Learned that opening/closing issues helps inform teammates about progress.

Timeline of Significant Events

- Physics engine blocked (Sprint 1-2) - Re-distributed tasks to put more focus on the engine.
- Pivoted authoring environment to use splines instead of a grid (mid Sprint 3).
- Split game file schema (Sprint 3).
- Cheat system integration (late Sprint 3).

Learnings from Managing a Large Project

- Emilio: Coordinating large groups of people with different schedules is hard.
- Georgiy: Important to solidify APIs early so that depending code doesn't break upon changes.
- Iurii: Merge conflicts are a real pain - sometimes features get broken and it takes a long time to get back on track.
- Mason: Keeping the design updated and scope clear made integration easier.
- Michael: How to manage many branches at once (delete deprecated once done)
- Paulina: In subgroups, everyone understands what they're working on, but it's important to inform each other and coordinate.

Learning from Creating a Positive Team Culture

- Emilio: If everyone's enjoying what they are creating and are willing to help each other, work feels fun.
- Georgiy: Asking for help and communicating any kind of progress should be encouraged.
- Iurii: When people show up proactively and are excited, work gets done much quicker
- Mason: Coming prepared to team meetings keeps discussions focused.
- Michael: Having hard design discussions first leads to clearer planning and scope later.
- Paulina: Feel a sense of ownership over the project so that you're more motivated

Team Contract Revisited

What worked:

- Discord worked (individual channels for splitting functionalities up).
- Giving frequent updates on who's doing what so that order of operations are ensured.

What could be changed

- Properly discuss certain design specifications with a teammate before implementation.
- Ask about updates more frequently, less of 'operating in the dark'.
- Setting 'mini-sprints' (daily standups) so that progress is more incremental.

Learning from Communication + Handling Negative Situations

- Emilio: Always being patient and understanding is more important than being upset, even if you disagree.
- Georgiy: An understanding tone invites people to work together.
- Iurii: Communication is key. It is important to be respectful to one another and be ready to help even if you are upset.
- Mason: Raising concerns early kept small issues from becoming bigger ones.
- Michael: Trusting that teammates will deliver their part so you can develop concurrently.
- Paulina: Being patient with one another and stepping up for your teammates if needed.

Thank you!

Any questions?